together that have the same value. This allows the statistics to be calculated by working with a few groups, rather than a large number of individual samples. Using this approach, the mean and standard deviation are calculated from the histogram by the equations:

EQUATION 2-6
Calculation of the mean from the histogram. This can be viewed as combining all samples having the same value into groups, and then using Eq. 2-1 on each group.

$$\mu = \frac{1}{N} \sum_{i=0}^{M-1} i H_i$$

EQUATION 2-7
Calculation of the standard deviation from the histogram. This is the same concept as Eq. 2-2, except that all samples having the same value are operated on at once.

$$\sigma^2 = \frac{1}{N-1} \sum_{i=0}^{M-1} (i - \mu)^2 H_i$$

Table 2-3 contains a program for calculating the histogram, mean, and standard deviation using these equations. Calculation of the histogram is very fast, since it only requires indexing and incrementing. In comparison,

```
100 'CALCULATION OF THE HISTOGRAM, MEAN, AND STANDARD DEVIATION
110 '
120 DIM X%[25000]                 'X%[0] to X%[25000] holds the signal being processed
130 DIM H%[255]                   'H%[0] to H%[255] holds the histogram
140 N% = 25001                    'Set the number of points in the signal
150 '
160 FOR I% = 0 TO 255             'Zero the histogram, so it can be used as an accumulator
170   H%[I%] = 0
180 NEXT I%
190 '
200 GOSUB XXXX                    'Mythical subroutine that loads the signal into X%[ ]
210 '
220 FOR I% = 0 TO 25000 'Calculate the histogram for 25001 points
230   H%[ X%[I%] ] = H%[ X%[I%] ] + 1
240 NEXT I%
250 '
260 MEAN = 0                      'Calculate the mean via Eq. 2-6
270 FOR I% = 0 TO 255
280   MEAN = MEAN + I% * H%[I%]
290 NEXT I%
300 MEAN = MEAN / N%
310 '
320 VARIANCE = 0                  'Calculate the standard deviation via Eq. 2-7
330 FOR I% = 0 TO 255
340   VARIANCE = VARIANCE + H[I%] * (I%-MEAN)^2
350 NEXT I%
360 VARIANCE = VARIANCE / (N%-1)
370 SD = SQR(VARIANCE)
380 '
390 PRINT  MEAN  SD               'Print the calculated mean and standard deviation.
400 '
410 END
```

TABLE 2-3